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A Comparative Intelligent Framework for Enterprise Cyber Threat Detection Using Extreme Learning Machine Across Neuro-Fuzzy, Fuzzy Logic, Isolation-Based, and Statistical Modeling Approaches

SPIC2A01-Project 2 for Applied Machine Learning

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ABSTRACT

- Enterprise cybersecurity requires efficient detection of dynamic and high-volume network threats.
- This work presents a comparative framework using Extreme Learning Machine across multiple detection models.
- Models compared include ANFIS, Fuzzy Logic, Isolation Forest, and Statistical Z-Score.
- Performance evaluation is done using repeated validation and statistical analysis methods.
- Results show that Extreme Learning Machine provides higher accuracy, better stability, and faster computation.



INTRODUCTION

- ❑ Enterprise networks generate large volumes of data, making threat detection complex and challenging.
- ❑ Traditional rule-based systems are less effective against evolving and dynamic cyber attacks.
- ❑ Machine learning techniques improve detection accuracy through automated pattern recognition.
- ❑ Extreme Learning Machine provides fast learning and efficient classification in high-dimensional data.
- ❑ Comparative analysis helps identify the most reliable model for enterprise cybersecurity applications.



OBJECTIVES

- To develop an enterprise cyber threat detection framework using Extreme Learning Machine.
- To compare ELM with ANFIS, Fuzzy Logic, Isolation Forest, and Z-Score models.
- To evaluate model performance using accuracy and statistical analysis.
- To analyze detection efficiency across intrusion, anomaly, and security event datasets.
- To identify the most reliable and scalable model for enterprise cybersecurity applications.



ALGORITHMS USED

- Extreme Learning Machine (ELM):

Fast learning neural network with random weights and high generalization capability for classification tasks.

- Adaptive Neuro-Fuzzy Inference System (ANFIS):

Combines neural networks and fuzzy logic for rule-based learning and decision making.

- Rule-Based Fuzzy Logic:

Uses predefined rules and membership functions to handle uncertainty in network data.

- Isolation Forest:

Anomaly detection algorithm that isolates outliers using random partitioning.

- Statistical Z-Score Model:

Detects anomalies based on deviation from mean using statistical thresholding.



METHODOLOGY

- Data preprocessing performed to remove missing values and normalize network features.
- Feature selection applied to identify important attributes for threat detection.
- Enterprise intrusion datasets used for training and testing the models.
- Extreme Learning Machine and comparison models trained over multiple iterations.
- Performance evaluated using accuracy and statistical analysis (SPSS).
- Comparative analysis conducted to determine the most effective model for cybersecurity tasks.



RESULT



- All proposed models demonstrated effective cyber threat detection capability across different enterprise scenarios.
- Extreme Learning Machine consistently achieved higher accuracy and better performance across all comparative models.
- Fuzzy-based and statistical models showed stable results but lower efficiency in handling large-scale data.
- Isolation-based and rule-based approaches performed adequately but lacked scalability and precision.
- Statistical evaluation confirmed that the performance differences between models are significant and reliable.

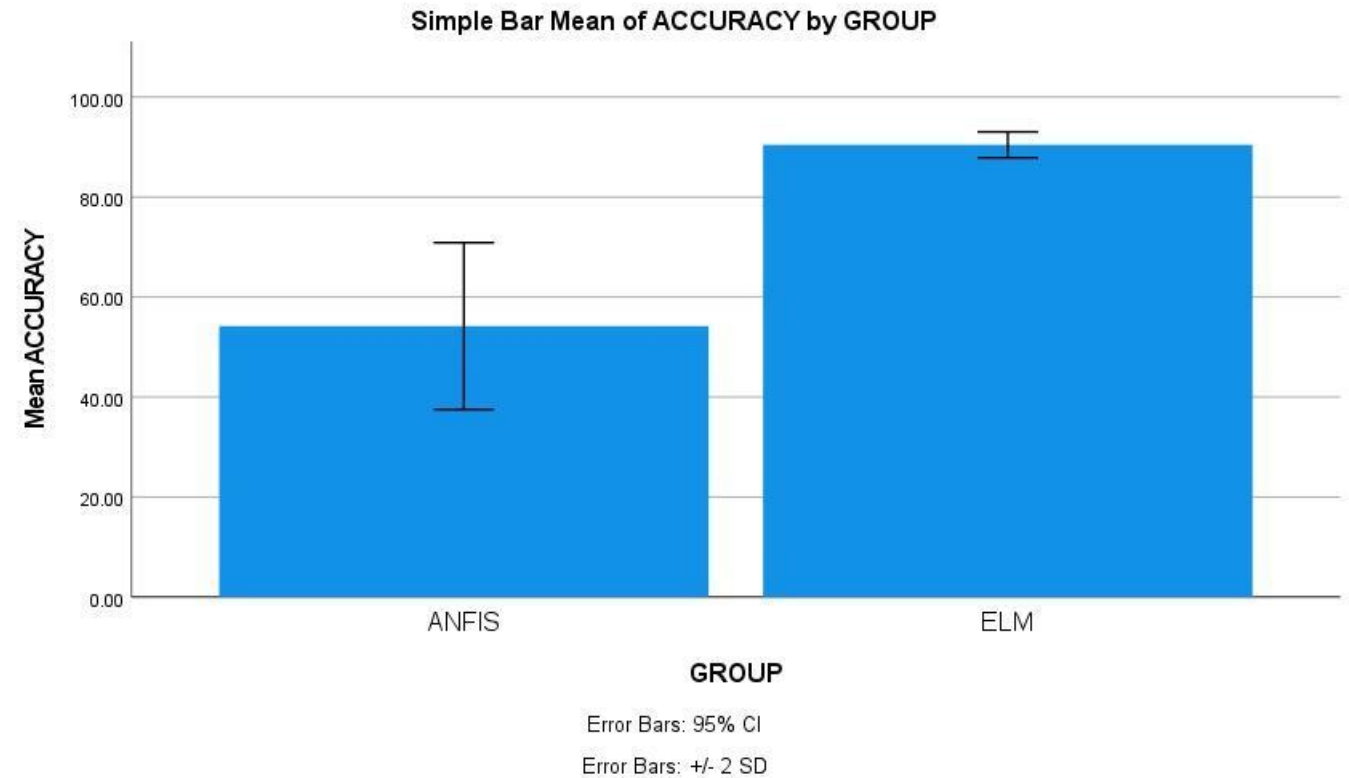


RESULT



Result 1: ELM vs ANFIS

- The SPSS analysis compares Extreme Learning Machine and Adaptive Neuro-Fuzzy Inference System for intrusion detection.
- Extreme Learning Machine achieved higher accuracy (90.4667%) compared to ANFIS (54.2000%).
- The independent samples test shows a statistically significant difference ($p < 0.001$).
- This indicates that ELM provides better classification performance.



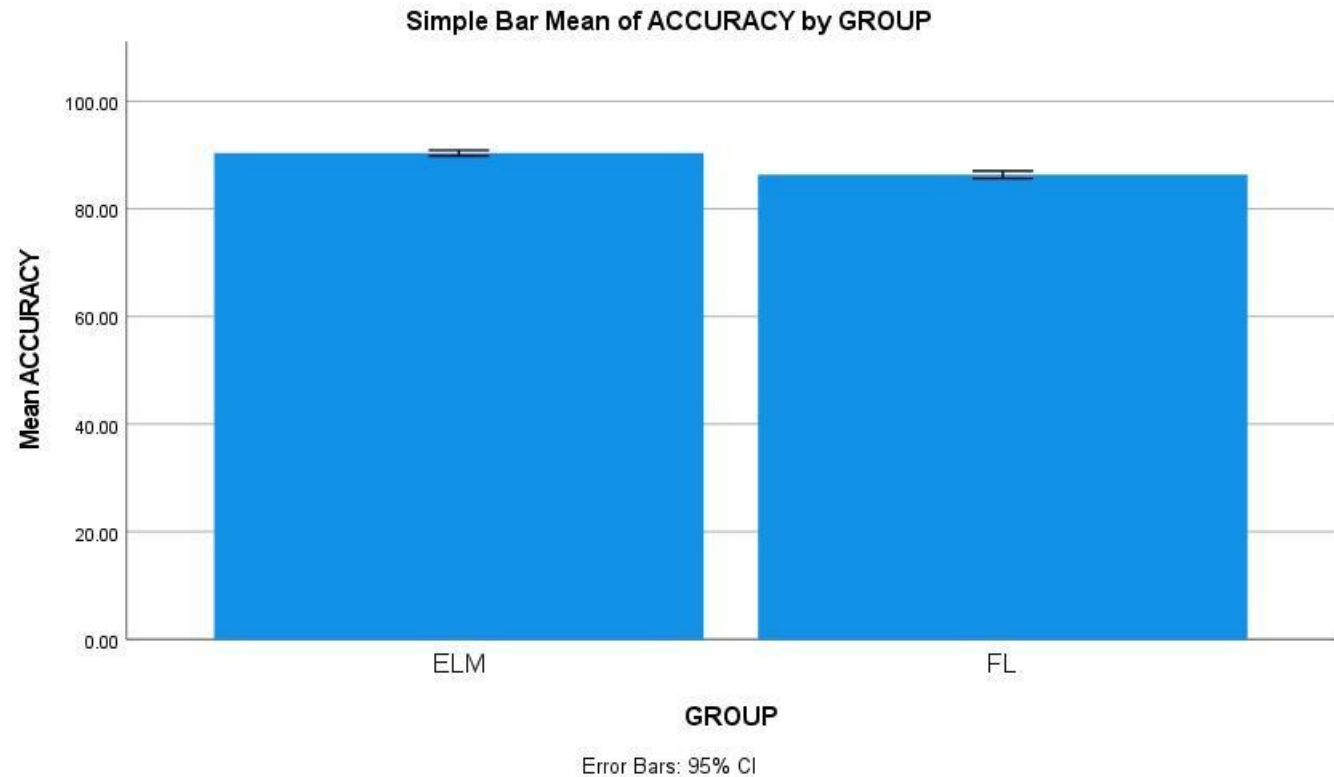


RESULT



Result 2: ELM vs Fuzzy Logic

- The comparison evaluates Extreme Learning Machine against Rule-Based Fuzzy Logic.
- ELM achieved higher accuracy (90.3833%) compared to Fuzzy Logic (86.3667%).
- Statistical analysis shows significant improvement ($p < 0.001$).
- This suggests that ELM improves detection reliability.



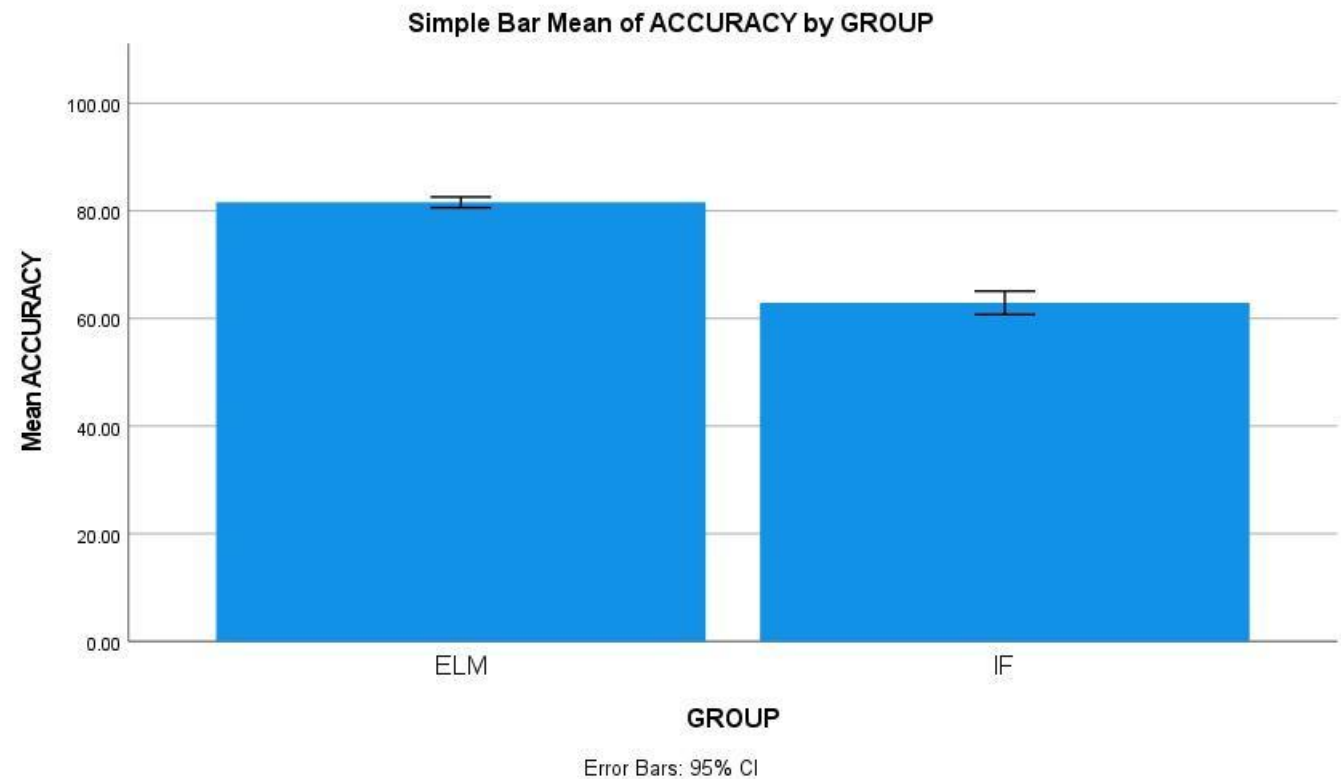


RESULT



Result 3: ELM vs Isolation Forest

- The SPSS results compare Extreme Learning Machine with Isolation Forest for anomaly detection.
- ELM achieved higher accuracy (81.6000%) compared to Isolation Forest (62.9074%).
- The statistical test shows significant difference ($p < 0.001$).
- This indicates better performance in high-volume data.



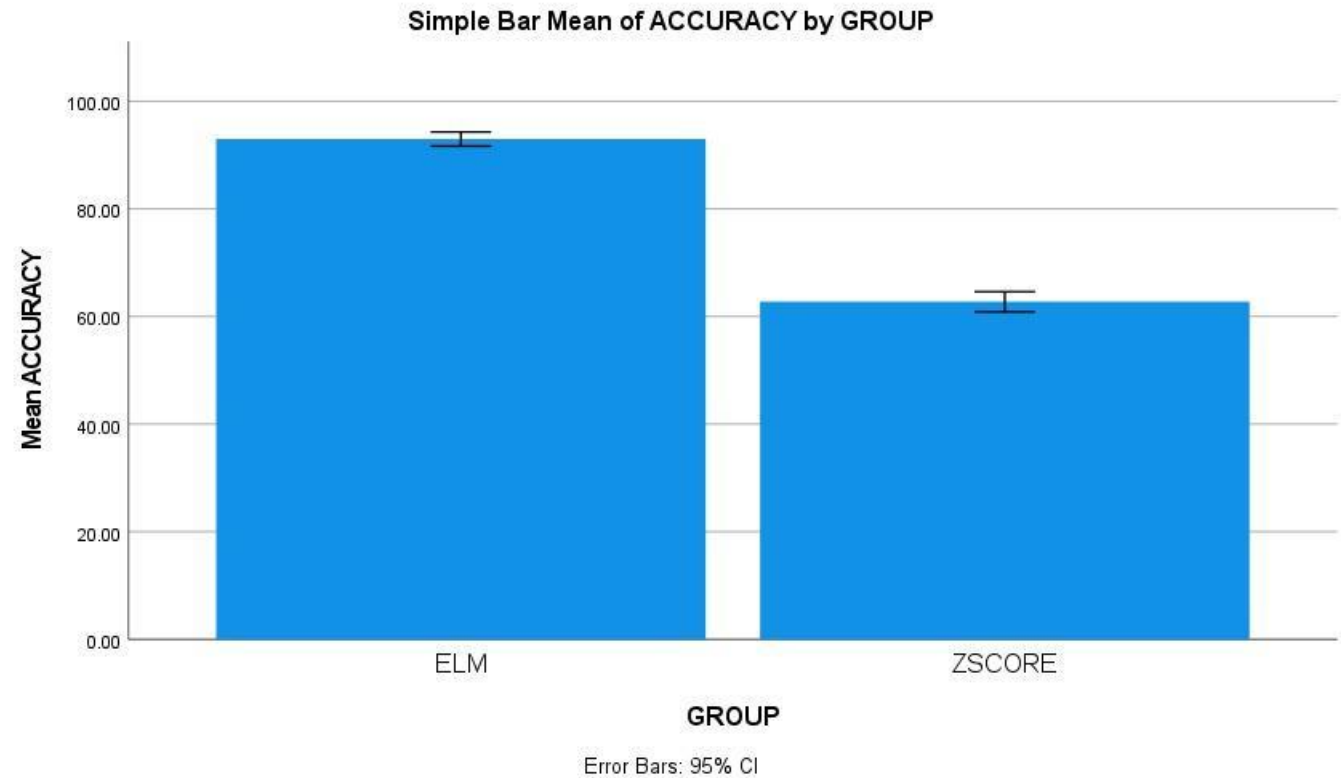


RESULT



Result 4: ELM vs Z-Score

- The analysis compares Extreme Learning Machine with Statistical Z-Score model.
- ELM achieved higher accuracy (92.9833%) compared to Z-Score (62.7500%).
- The independent samples test shows strong significance ($p < 0.001$).
- This highlights improved threat detection capability.





LIMITATIONS



- Performance depends on the quality and diversity of the dataset used.
- Extreme Learning Machine requires proper selection of hidden neurons for optimal results.
- Rule-based and statistical models show reduced performance in large-scale and complex data.
- Advanced models may require higher computational resources and tuning.
- Results may vary when applied to real-time or unseen enterprise network environments.



FUTURE SCOPE

- Extend the framework for real-time enterprise cyber threat detection systems.
- Use larger and continuously updated datasets to improve model generalization.
- Develop hybrid models by combining ELM with other intelligent techniques.
- Integrate explainable AI methods for better interpretability of results.
- Deploy the system in cloud and large-scale enterprise environments for practical applications.



CONCLUSION

- The study demonstrates that machine learning significantly improves enterprise cyber threat detection.
- Extreme Learning Machine consistently achieved higher accuracy across all comparisons.
- ELM showed better performance in handling high-dimensional and large-scale network data.
- Comparative analysis confirms ELM as a reliable and scalable model for cybersecurity applications.
- The proposed framework provides an effective solution for real-world enterprise threat detection systems.



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THANK YOU !